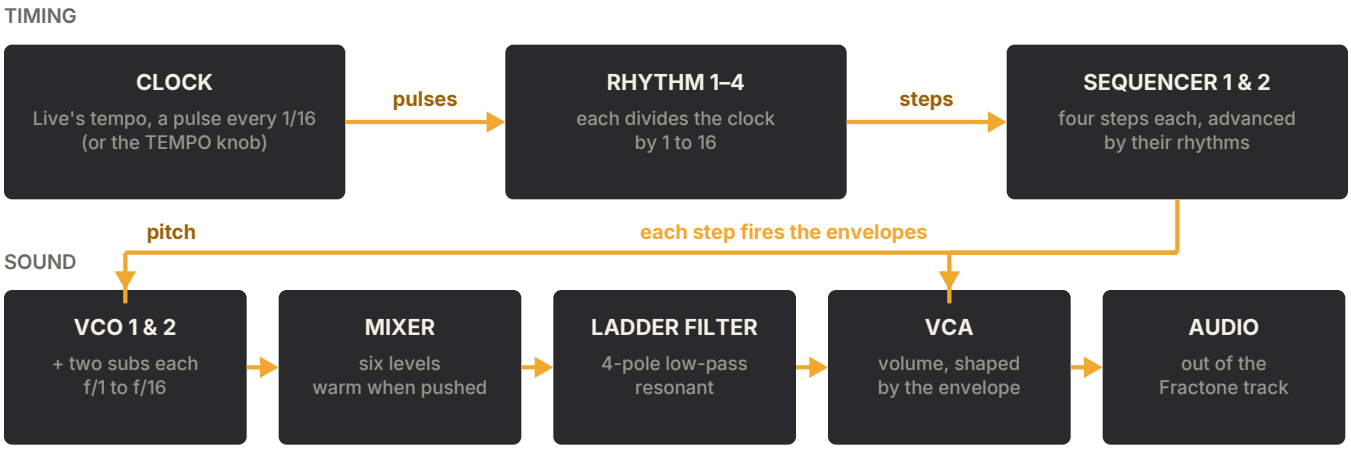
FRACTONE

f/1 to f/16: the undertone series

Getting started: your first polyrhythm in Ableton Live

Fractone is a polyrhythmic undertone synthesizer. Two oscillators each carry two subharmonics, the note divided by 1 to 16, and four rhythm generators divide one clock by 1 to 16 to drive two four-step sequencers. It makes its own sound, so one MIDI track in Live is all it needs. This guide gets it playing in a couple of minutes, then builds your first polyrhythm one step at a time.



How a note happens: the clock drives the rhythms, the rhythms advance the sequencers, and the sequencers set the pitch and fire the envelopes.

You'll need

- Ableton Live 11 or 12, any edition that can load VST3 or Audio Unit plug-ins.
- Fractone installed. Builds are on the GitHub releases page; on macOS **Fractone.vst3** goes in **~/Library/Audio/Plug-Ins/VST3** and **Fractone.component** in **~/Library/Audio/Plug-Ins/Components**. Building from source installs both for you.
- Nothing else: no second track and no other instrument.

> **Coming from Dicey?** Fractone works the other way round. It makes sound and listens to MIDI, so there's no routing to set up: the track it sits on is the track you hear.

Set up

1 Let Live see Fractone

Open **Live → Settings** (⌘,) and choose the **Plug-Ins** tab. Turn on **Use VST3 Plug-in System Folders** (or **Use Audio Units**), then press **Rescan**.

In the browser, open **Plug-Ins**. Fractone is under **Taddei**.

2 Put it on a MIDI track

Insert a MIDI track (**Create → Insert MIDI Track**, or ⌘⇧T) and drag **Fractone** onto it. It opens on the **Init** patch: VCO 1 and its first subharmonic, played by Sequencer 1.

3 Press play

Fractone follows Live's tempo and transport. Press Live's play button and you'll hear a four-note phrase, while the step LEDs walk across Sequencer 1. The status display on the bottom rail reads **SYNC 120.0 BPM 1/16 RUNNING**.

Stop Live and Fractone stops too. Starting again, jumping or looping resets the sequencers and rhythms, so every pass begins on the same downbeat.

> **Nothing?** The display says **WAITING FOR HOST** until Live is playing. If it says **PLAY OFF**, click **PLAY** on the panel. Troubleshooting on the last page has the full list.

How the clock works

Each clock pulse is a 1/16 of Live's tempo (**RATE** on the bottom rail changes it, from 1/1 to 1/64). A rhythm generator counts those pulses: Rhythm 1 in the Init patch is set to 1/2, so it fires on every second pulse and Sequencer 1 moves in eighth notes. Turn the RHYTHM knob clockwise for faster, counter-clockwise for slower; hover it to read the division.

Turn **SYNC** off to run from the panel's own **TEMPO** knob instead, one pulse per beat from 20 to 3000 BPM, started and stopped with PLAY.

Your first polyrhythm

Start from Init, with Live playing. Each step adds one idea.

1 Tune the steps

Turn Sequencer 1's **STEP** knobs. Centre leaves the pitch alone; clockwise goes up and counter-clockwise down, across the **SEQ OCT** range (±2 octaves in Init). With **QUANTIZE** on 12-ET each knob snaps to semitones. Or click **DICE**: step 1 stays put, and steps 2–4 land on consonant intervals from it.

2 Add an undertone

Turn up **SUB 2 LEVEL** under VCO 1. Its **SUB 2 FREQ** knob divides VCO 1's pitch: fully clockwise is unison, then 1/2 an octave down, 1/3 an octave and a fifth down, 1/4 two octaves, 1/5 two octaves and a major third. The sub follows every note the sequencer plays, in tune by construction.

3 Add a second rhythm

Light **SEQ 1** under **RHYTHM 2** (set to 1/3). Sequencer 1 now advances on every second pulse and on every third. The pattern repeats every six pulses, but the gaps between steps are uneven, and that unevenness is the groove. Press **RESET** to line both up again.

4 Bring in the second voice

Turn up **VCO 2 LEVEL**, light **SEQ 2** under **RHYTHM 4**, and turn RHYTHM 4 to 1/5. Then click Sequencer 2's **DICE**. Sequencer 2 plays VCO 2 (**OSC 2** is lit under SEQ 2 ASSIGN) on its own rhythm, and with 1/2, 1/3 and 1/5 running together the whole pattern takes 30 pulses to come back around.

5 Shape it

CUTOFF and **RESONANCE** set the tone, and **VCF EG AMT** decides how far each step opens the filter. **VCA DECAY** sets the note length: short for plucks, long to let notes ring into each other.

6 Save it

Click **SAVE** on the bottom rail and give the patch a name. Presets keep every knob, switch and patch cable, and show up in every Live set.

> **Tuning by ear** Hold **EG** until it blinks. The envelopes stay open, so you hear a steady tone. With Live stopped, **NEXT** steps through the sequence while you tune each knob. Press EG again to go back.

Find your way around



1	**Sequencers**	Two four-step sequencers. Sequencer 1 plays VCO 1, Sequencer 2 plays VCO 2. The LED shows the current step.
2	**DICE**	Keeps step 1 and rolls steps 2–4 to consonant intervals from it. MIDI-mappable as Dice Seq 1 and 2.
3	**Polyrhythm**	Four rhythm generators, each dividing the clock by 1 (clockwise) to 16. Their SEQ buttons pick which sequencer they advance.
4	**Transport**	TEMPO (free-running only), RESET, EG, NEXT, PLAY and TRIGGER.
5	**Oscillators**	VCO FREQ (four octaves up from middle C), the WAVE switch, and SUB FREQ dividing each oscillator by 1 to 16.
6	**SEQ OCT**	The step knobs' range: ±1, ±2 or ±5 octaves.
7	**SEQ ASSIGN**	What each sequencer moves: the oscillator's pitch, a sub's divisor, or any mix of the three.
8	**Mixer**	Levels for all six sources. Push several high for warm saturation.
9	**QUANTIZE**	Off, 12-ET, 8-ET, 12-JI or 8-JI, for the FREQ knobs and the steps.
10	**Filter and VCA**	CUTOFF, RESONANCE (rings on its own at maximum), VCF EG AMT (centre is none) and VOLUME.
11	**Envelopes**	Attack and decay for the filter and the VCA, shared by every source.
12	**Patchbay**	17 inputs (plain labels) and 15 outputs (white labels). Drag cables between them.
13	**Bottom rail**	Presets and SAVE, SYNC and RATE, LOGIC, FINE TUNE, DRIFT, MIDI channel and the status display.

> **Hold for more** Three buttons do something extra when held. **EG** holds the envelopes open. **QUANTIZE**, held until its LED blinks, sends unquantized CV from the SEQ jacks. **SEQ OCT**, held until its LED blinks, keeps the SEQ jacks at ±5 V.

Patch, play and record

Patching

Drag from an output (white label) to an input to add a cable. Grab a plugged input to move its cable, and right-click a jack to unplug it. **CLEAR CABLES** on the bottom rail removes them all. Three to try:

SEQ 2 → CUTOFF	Sequencer 2 plays the filter as well as VCO 2. Unassign OSC 2 to make it a filter sequence only.
CLOCK → VCA	Every clock pulse opens the VCA: a ticking pulse under everything. The Odd Rock preset uses it.
SEQ 2 → RHYTHM 1	With RHYTHM 1 at noon, each Sequencer 2 step sets how fast Rhythm 1 runs. Hold SEQ OCT until it blinks for the widest range.

Presets

Click the preset name on the bottom rail to browse eleven factory patches, then any you've saved. **Ladder Kick** uses no oscillators at all: the self-oscillating filter is the drum. An asterisk after the name means you've changed something since loading it.

Play it from a clip

MIDI notes on Fractone's track transpose both oscillators relative to C3 (MIDI note 60) and open the envelopes, so a clip can move the running pattern into a new key, or play Fractone as a monosynth with the sequencers off.

Put DICE on a controller

Click the device's **Configure** button, click a DICE button so its parameter appears, then use MIDI Map mode (**⌘M**) and press a pad. Map it as a momentary button. Draw a short pulse on the Dice Seq 1 automation lane every few bars and the phrase re-rolls as the song plays.

Record what it plays

Arming Fractone's own track records MIDI, and Fractone plays itself, so that clip stays empty. To capture the sound:

- Create an audio track and set its **Audio From** to the Fractone track, **Post FX**.
- Set its **Monitor** to **Off**, arm it, and record. Turn on **Automation Arm** to keep your knob moves as well.

Troubleshooting

Fractone isn't in the browser	Turn on **Use VST3 Plug-in System Folders** (or **Use Audio Units**) and press **Rescan**. A build downloaded on macOS needs its quarantine flag cleared first; the README's Download section has the command.
The display says WAITING FOR HOST	Press Live's play button. With SYNC on, Fractone only runs while Live does.
Running, but no sound	Check **VOLUME** and the mixer LEVEL knobs. If **EG** is unlit, steps don't fire the envelopes; click it, or hold it until it blinks to hold them open.
A sequencer doesn't move	It needs at least one rhythm with its **SEQ** button lit. Holding RESET, or a cable into the RESET jack, also holds step 1.
Everything pulses together	That's the design: one filter and one VCA shared by all six sources. For a drone under a pulse, use two Fractone tracks, one with EG held.
Recording captures nothing	The armed MIDI track records MIDI. Record onto an audio track with **Audio From** set to the Fractone track.
Dice Seq 1 isn't in Live's MIDI map	Click **Configure** on the device, then click that DICE button so the parameter appears.
The preset name has an asterisk	You've changed something since loading it. **SAVE** keeps the change.

Everything else, control by control, is in the README.